



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath

The University of Dublin

School of Computer Science and Statistics

What course should I study? A personalised Recommender System for Irish college courses

Brendan McCann

Supervisor: Dr Owen Conlan

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Declaration

I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at <http://www.tcd.ie/calendar>.

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1 Research Topic Summary

1.1 Motivation for Research Topic

"What course should I study?" is a daunting question to people for many reasons. For instance, the course an individual pursues could make a huge difference to the rest of their life. Moreover, the decision comes with personally confronting questions such as: *"What do I want to do with the rest of my life?"*, *"What do I love doing?"* and *"What am I good at?"*, among many others. In Ireland, most second-level students are faced with making that decision at around 17 years old, an age with limited life experience and knowledge about the world.

In choosing a college course, an individual is also influenced by a multitude of external factors. Parents are shown to be one of the most influential factors on an individual's college course choice [1]. An individual's gender, socioeconomic status and their family's educational background are among the most prominent societal influences on an individual's college course choice [1].

The difficulty in choosing a suitable college course is the primary motivation for this research topic.

1.2 Research Question

The main research question that will be examined in this dissertation is: *How can a Recommender System assist individuals in finding a suitable Irish college course?*

1.2.1 List of Research Objectives

1. Investigate what data should be gathered from users to recommend suitable college courses.
2. Create a Recommender System (RS) to suggest personalised Irish college courses.

3. Measure the user's satisfaction and performance of the RS using established evaluation techniques.

1.2.2 Approach/Methods Planned to Achieve Research Objectives

1. Conduct multiple interviews with secondary school Guidance Counsellors to further the understanding behind college course choice.
2. Create multiple prototypes of an RS and continuously optimize towards a stronger solution.
3. Investigate the established benchmarks and techniques for evaluating RS performance.

1.2.3 Summary of Evaluation

RS's are designed to be used by humans. Following from this, many of the evaluation metrics for RS's require human input. For instance, the novelty of recommendations is one such metric, and this would have to be measured subjectively by the user. Thus, I am planning on recruiting human participants for a study.

In RS's, it is common to use a baseline model. In our case, the baseline model would produce a list of the most popular college courses. In the study, the user will interact with the system and answer the prompted questions. The user will then be presented with two lists of college course recommendations: the baseline model's recommendations, and the actual recommendations. This allows us to compare our RS's performance with a baseline model.

As mentioned, there are many evaluation metrics of an RS's performance. These include: relevance, diversity, novelty, serendipity and trust. There are established and clearly documented methods to obtaining these metrics. These metrics require user input from the study, and they will be instrumental in evaluating the overall performance and effectiveness of the RS.

1.2.4 Anticipated Contributions

Many similar papers have a heavy focus on the technical design of their solutions, and the background reading is weak. In this paper, I hope to synthesise comprehensive background knowledge with appropriate technical solutions. I anticipate this research will contribute to a more wholistic solution in the intersection between vocational psychology and artificial intelligence.

As well as this, this research has real-world interest. One of the Guidance Counsellors even expressed interest in transforming this idea into a commercial product. This demonstrates the viability of this research beyond an academic context.

2 Literature Review Plan Summary

Concepts for literature review:

- (a) Theories of college/career choice.
- (b) Theories of job/career satisfaction.
- (c) Understanding the motivations and external influences behind an individual's college course selection.
- (d) Investigating any AI-inspired solutions to this problem.

2.1 Keyword Search Summary

Keywords: Recommender Systems, Machine Learning, Large Language Models, Guidance Counselling, College Course, Career, Job, University, Vocational Psychology

2.1.1 Keyword Search Results

Seven references resulting from the keyword search, and their anticipated relevance, are:

- (i) [2]: relevant, because it investigates the utility of guidance counselling in university contexts.
- (ii) [3]: relevant, because it discusses an approach for guidance counselling to university students.
- (iii) [4]: relevant, because it does a survey of AI-powered solutions for career counselling.
- (iv) [5]: relevant, because it uses an LLM-powered RS for vocational training programs for students.
- (v) [6]: relevant, because it uses a conversational agent for career guidance.
- (vi) [7]: relevant, because it uses an AI-powered career recommendation system.
- (vii) [8]: relevant, because it uses a deep learning-powered RS for career guidance.

2.1.2 Keyword Search Approach

The search engine used was **Google Scholar**. Two searches were performed:

1. **Without** the Computer Science concepts: *"Guidance Counselling, College Course, Career, Job, University, Vocational Psychology"*
2. **With** the Computer Science concepts: *"RS's, Machine Learning, Large Language Models, Guidance Counselling, College Course, Career, Job, University, Vocational Psychology"*

I chose to do two separate searches, as I also wanted some papers that focused on vocational psychology without a technical lens. This allowed me to enhance my background understanding of the problem.

During my search, there was one paper [9] that utilised cognitive analytics to understand students personality tests, which would then be used to further understand their professional choices. While I can see the semantic similarity with my research, this research was not entirely relevant, so I chose to exclude it from the list.

With all papers, I performed a quick background on the author and the journal. This check included checking the author's h-index and the legitimacy of the journal the paper was published in.

2.2 AI Search Summary

2.2.1 AI Search Results

Seven references resulting from the keyword search, and their anticipated relevance, are:

- (i) [10]: relevant because it investigates the psychology behind RS's.
- (ii) [11]: relevant because it uses an AI-powered career to mitigate anxiety with students when making college course decisions.
- (iii) [12]: relevant because it does a systematic review of Machine Learning applications in the field of Higher Education.
- (iv) [13]: relevant because it proposes an RS for student academic advising.
- (v) [14]: relevant because it evaluates the performance of Large Language Models in the realm of college course recommendations.
- (vi) [15]: relevant because it also evaluates the performance of Large Language Models in the realm of college course recommendations.

(vii) [16]: relevant because it links learning objectives and career competencies.

2.2.2 AI Search Approach

The AI model used was **Gemini 3 Pro**. The prompt used:

*Act as an expert in the literature of RS's and Vocational Psychology. I am currently working on my masters thesis. My project seeks to create an RS that suggests college courses to individuals. I am currently doing a literature review, and I would like you to recommend **8 scientific papers** that are relevant based on the keywords I will supply to you here: "RS's, Machine Learning, Large Language Models, Guidance Counselling, College Course, Career, Job, University, Vocational Psychology"*

At first, all I put in the prompt was the keywords. Afterwards, I iterated the prompt to add more context, telling it about my masters thesis and what my research seeks to achieve. I also asked it to assume the role as a domain expert to enhance the quality of results.

I came across one paper which used an RS to recommend internships to students [17]. I do see the similarities between this and my current research, but I chose to exclude it from the list because it matches internships and not college courses.

2.3 Comparison/Reflection

There were many similarities between the AI and keyword search results. When it came to finding papers lying in the intersection between RS's and college courses, both searches returned papers that were similar in content and legitimacy. Furthermore, when I wanted to find papers that just focused on college courses and had nothing to do with Computer Science, I found that both the AI and keyword search results to have papers with similar relevance, utility and legitimacy. I did not find a considerable advantage to using AI search in this case.

Interestingly, before the AI search recommended papers that lay at the intersection of RS's and college courses, it first suggested papers regarding the human psychology behind RS's [10]. Although this was not explicitly related to my specific research intersection, I found this paper to be very insightful and useful.

It is my opinion that AI search should not replace keyword searches. My findings is that they both provide papers with similar relevance, utility and legitimacy. That being said, I believe AI search can complement keyword search because it can provide papers with a level of serendipity - an unexpected, yet useful paper suggestion.

3 Supporting Information

3.1 Research Ethics Considerations

RS's are designed to be used by human users. In order to effectively measure RS's, they would need human participants. I am planning on conducting a study involving human participants, where they would interact with the RS. In order to suggest relevant courses to the user, the RS must process some personal data from the user.

Seeing as my research involves humans and processing of their personal data, my research requires a Research Ethics Application.

3.2 Technology Ethics Considerations

The main stakeholder at risk is the human user of the RS. Choosing a college course is a nuanced and personal decision, and the RS I am designing does not entirely capture this nuance. With the growing popularity of Large Language Models (LLMs), some humans are becoming lazier by offhanding their critical thinking to LLMs. [18]

My primary ethical concern is that users could grow an over-dependence on the tool, potentially reaching a stage where the user lets the tool make the college course decision for them. This is not my intention with this research. My primary intention with this research is to design a tool would augment the difficult decision-making process behind choosing a suitable college course, not to replace it.

3.3 Skills you aim to Acquire through Dissertation Research

3.3.1 Research Skills

In terms of research skills, these can be divided into two categories:

Academic Writing

I aim to develop my ability to...

1. Write academically, i.e. objective, impartial and professional.
2. Write with rigour, i.e. crafting reasonable and convincing arguments.
3. Build upon existing scientific knowledge by citing relevant papers appropriately.
4. Write coherently, i.e. to have flow between my arguments.

Academic Reading

I aim to develop my ability to...

1. Search for academic papers that would be relevant and useful for my research.
2. Maintain a database the papers I've read, what I've learned from these papers, and the further questions I have.
3. Read and understand academic papers. This can be mentally exhausting, especially as someone who is not familiar with the academic style of writing.

3.3.2 Technical Skills

1. Machine Learning (ML). Most RS's use ML in their decision-making process. I will get my hands on different ML models and further my understanding of ML in practice.
2. How to design an effective RS. Good ML alone does not guarantee a good RS. I will have to make design decisions in order to make diverse, relevant and novel recommendations to the user.

3.4 Workplan

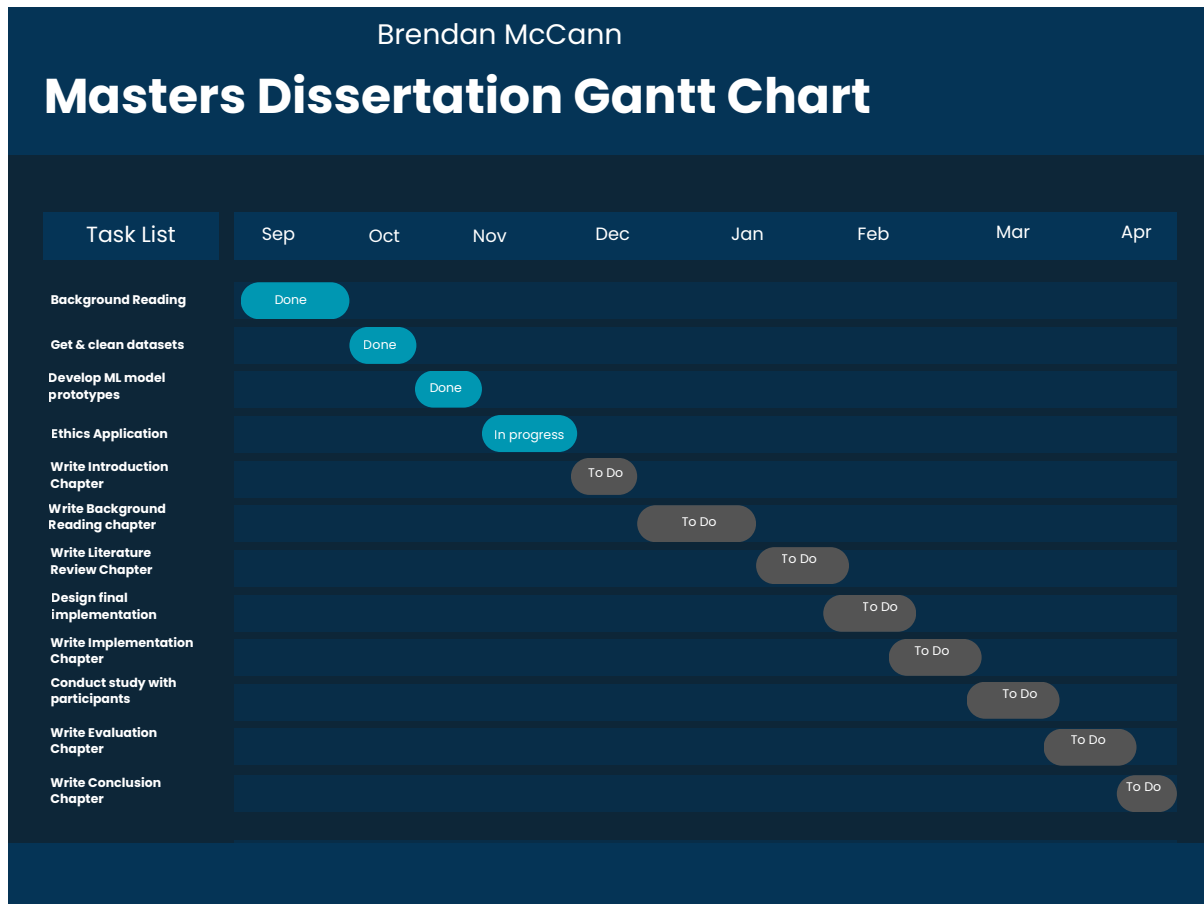


Figure 3.1: Gantt chart of dissertation project broken down into steps from start of semester 1 to submission date

3.5 Pivots in Research Plan

This research plan was shared with my supervisor to seek feedback. I found his opinions to be quite insightful, and in some cases, unexpected. I will describe my three pivots below.

Firstly, my supervisor noted that my technology ethics section did not have enough coverage. In the case of participants taking part in my study, it should be considered that some of the participants have committed themselves, in some shape or form, to a college course or vocation. Through interacting with the Recommender System, they may discover a college course that they wish they would have studied. This poses possibilities of regret or distress for the participant. This was an important concern to raise, and this consideration will be included in my ethics application.

Secondly, my supervisor remarked that I should also explore academic papers which have not

leveraged LLM's. For instance, he suggested I search for papers that exclusively use ML to suggest college courses. His main point of concern was that I was searching down a lens that was too narrow, and potentially missing other vital pieces of research in my particular research niche.

Lastly, my supervisor observed my Gantt chart. He mentioned that in many cases, a particular task is not "done" until the final deadline of the thesis. For example, he mentioned that even if I write up the entire Introduction Chapter in January, it may still face significant edits in March or April. He stressed that I should be open to flexibility, and not to enforce unnecessary rigidity in my project planning. While this is not strictly a "pivot", I found this advice to be insightful.

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